

India-first Business Models for Adaptive Gantry and VLA Lab Automation in Diagnostic Labs

Executive summary

The highest-probability commercial model for an India-first adaptive gantry plus VLA lab-automation startup serving mid-to-large diagnostic labs is a **hybrid model**: charge an upfront deployment fee for hardware and integration, lock in recurring software and support revenue, and add smaller recurring revenue from workflow packs, PM kits, and SLA tiers. This structure best matches the underlying problem described in your validation materials: Indian labs have real pain in manual “glue work,” sample movement, routing, and repetitive handoffs, but the weakest part of the current concept is still business model design, unit economics, and GTM clarity. The pitch deck also points to a sensible phase-one wedge: gantry-based tube handling in constrained lab spaces, where fixed gantries carry lower safety and integration burden than general-purpose arms. [filecite:turn0file1](#) [filecite:turn0file2](#)

A pure **CapEx hardware sale** is viable, especially for private reference labs and chain hubs that already buy analyzers and automation, but it creates lumpy revenue and weak recurring economics. A pure **RaaS or Opex subscription** lowers adoption friction and works well for lighthouse pilots, but it pushes asset-financing and utilization risk back onto the startup. **Transactional per-test pricing** and **revenue-share** both sound attractive to buyers, but they fit poorly when the startup does not own assay chemistry or reagent economics; for phase-one pre/post-analytical automation, those models often under-monetize value unless protected by minimum guarantees. **Licensing and IP** can be highly attractive later, especially for OEM or chain-wide software integrations, but it is not the best day-one monetization path. ¹

The market signal is real. India's large diagnostics networks are operating at national scale and still growing sample volumes. Dr Lal PathLabs says it has collected 85.6 million samples, operates 290-plus labs and 6,600-plus patient service centres; Reuters reported roughly 8.5% to 10% sample-volume growth across recent periods. Metropolis says it has 170-plus advanced labs, 12,000-plus patient touchpoints, 4,000-plus tests and profiles, and more than 2,000 scientific officers; recent commentary also points to continuing growth in specialty and preventive testing. In parallel, Indian hospitals are already buying or deploying high-end automation: SGPGIMS inaugurated a fully automated pathology laboratory processing more than 4,000 tests per hour, while AIIMS Bhopal deployed a cobas pro system processing more than 2,000 tests per hour under a reagent-rental structure worth over ₹3 crore. That is the central business-model implication: **buyers already understand automation value, but the incumbent price points are often too high, too centralized, and too rigid.** ²

For pilots, the best order of attack is: **regional private reference lab first, chain hub lab second, hospital enterprise or PPP/public lab third.** The recommended contract structure is a **paid pilot with explicit success metrics and a pre-agreed conversion path** into either hybrid purchase or selective RaaS. That avoids the dead zone of “free pilot” behavior, keeps the startup from fronting too much balance-sheet capital too early, and still gives procurement teams enough evidence to justify rollout.

[filecite:turn0file1](#) ³

Scope and modeling assumptions

This report assumes the startup's **phase-one product** is not a full chemistry analyzer or a full total-lab-automation track. Instead, it is a modular automation cell for **tube pickup, barcode reading, sorting, buffer handling, rack transfer, analyzer handoff, and related traceability-heavy “glue work”**. That matches the wedge described in your materials: gantries in constrained spaces, lower safety burden than arms, and an evolution path toward richer bench automation once operational data has been captured.

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The modeled **target customer** is a NABL-accredited or accreditation-seeking diagnostic lab doing roughly **800 to 5,000 samples per day**, with LIS connectivity, multiple analyzer benches, meaningful TAT pressure, and a quality team that cares about sample tracking, calibration, and auditability. Large Indian chains already signal those operating characteristics publicly: Metropolis highlights sample tracking, calibration, and internal and external quality checks, while Dr Lal highlights real-time QC monitoring, quality checks on every sample, and daily quality indicators. NABL's medical-laboratory accreditation framework is anchored to ISO 15189, which reinforces why auditability, validation, and traceable process changes matter commercially. ⁴

All INR conversions that anchor imported comparables use the rupee close reported by Reuters on **2026-06-04**, about **₹95.785 per US dollar**. At that rate, Opentrons' public list pricing implies a Flex configuration starting at about **₹25.3 lakh**, with workstation bundles ranging from roughly **₹51.9 lakh to ₹92.6 lakh**. Those figures are useful only as directional comparables, because a clinical-diagnostics deployment in India will typically need extra safety engineering, analyzer integrations, FAT or SAT style deployment work, IQ or OQ validation support, onsite training, spare inventory, and stronger uptime obligations than a research-lab robot. ⁵

The unit-economics model below is therefore an **explicit author model**, not a market quotation. It assumes a deployed direct-cost stack per cell of roughly **₹12 lakh to ₹18 lakh**, broken into industrial BOM, assembly and calibration, installation and site integration, and a first-year spare/service reserve, with software and cloud or monitoring cost of about **₹0.1 lakh to ₹0.4 lakh per month** depending on how much inference is done at the edge versus in the cloud. The labor-saving and TAT value model assumes the main economic win is not reagent margin but a combination of fewer manual touches, lower error or rerun burden, throughput elasticity, and better bench utilization. That assumption is important, because unlike analyzer vendors using reagent-rental structures, a gantry automation startup does **not** control test chemistry or assay menus. ⁶

Modeled direct-cost stack per deployed cell	Indicative range in INR
Industrial BOM including motion, structure, vision, controls, safety	₹9 lakh–₹13 lakh
Assembly, calibration, factory testing	₹1.5 lakh–₹2.5 lakh
Installation, FAT/SAT-style deployment, IQ/OQ-style validation support, training	₹1.5 lakh–₹3 lakh
First-year spare and field-service reserve	₹1 lakh–₹2.5 lakh

Modeled direct-cost stack per deployed cell	Indicative range in INR
Ongoing software, cloud, monitoring, and support operations	₹0.1 lakh–₹0.4 lakh per month

Modeled estimate by the author, anchored to public comparables and validation/support structures described by Opentrons, Automata, NVIDIA, and Indian automation deployments rather than vendor quotes for a specific build.

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Market trends and business model implications

The Indian diagnostics market is large enough, distributed enough, and quality-sensitive enough to support a specialized automation company. Dr Lal's public site shows national scale across labs and collection infrastructure, and Reuters has repeatedly reported high-single-digit to low-double-digit sample growth, driven in part by expansion into smaller cities and rising health awareness. Metropolis publicly positions itself as a pan-India network with advanced labs, large patient touchpoints, and deep scientific staffing, while recent market commentary links its performance to specialty and preventive testing. That combination matters because it raises both the operational load and the value of standardization across sample intake, routing, and analyzer handoff. 8

The business-model implication is that the startup should monetize around **workflow reliability and throughput**, not just "robot novelty." These labs already expose sample tracking and digital report workflows to end users, and they publicly emphasize calibration, proficiency, and QC. In other words, the buyer is not purchasing a fascinating robot; the buyer is purchasing **fewer failure points, tighter TAT, lower manual handling, and a cleaner quality narrative during accreditation or audit**. That is exactly why recurring software and service have to be part of the core offer, not an optional add-on. 9

There is also clear evidence that automation budgets already exist, but incumbent solutions often sit at a much higher price and complexity tier. SGPGIMS' fully automated pathology laboratory can process more than 4,000 tests per hour. AIIMS Bhopal's cobas pro deployment can process more than 2,000 tests per hour and was acquired under a reagent-rental arrangement worth more than ₹3 crore. At the lighter end of public-health infrastructure, Karnataka's PM-ABHIM-backed district integrated public health laboratory was established at about ₹1.1 crore. The practical takeaway is not that a startup should copy these systems; it is that buyers already understand the category, and a modular Indian product priced well below these benchmarks can be positioned as a **budget-responsible alternative for a narrower, high-friction workflow wedge**. 10

The competitive environment also favors automation that shortens TAT and hardens process discipline. Amazon's entry into home diagnostics in major cities signaled to public markets that digital platforms can pressure incumbent diagnostics players on convenience and patient experience. Meanwhile, both Dr Lal and Metropolis emphasize report access, tracking, and operational convenience on their public properties. That means backend lab automation is not merely an internal efficiency tool; it increasingly supports customer-facing experience and chain-level SLA performance. 11

Finally, the product concept in your own deck is commercially aligned with this market structure. Your problem-discovery work emphasizes manual "glue work," fragmented departments, high-cost consultant-driven status-quo automation, and underutilized staff, while the pitch deck argues that fixed gantries fit

constrained spaces and impose lower safety burden than mobile arms. Those are important business-model signals: the best first offer is one that can be sold as **targeted operational infrastructure**, not as a moonshot humanoid or a generalized robotics platform.

Comparative analysis of business models

The central commercial reality is this: **phase-one adaptive gantry automation is economically closer to workflow infrastructure than to assay chemistry**. That makes some familiar diagnostics business models less attractive than they first appear. Reagent-rental works for analyzer vendors because they capture recurring assay consumption. A startup focused on pre-analytical or inter-bench automation typically does not.

Model family	Revenue streams	Indicative pricing mechanics in INR	Unit economics and buyer payback	Cash-flow profile and capital need	Adoption, procurement, and regulatory constraints	Risk vectors
CapEx hardware sale	Hardware, installation, integration, AMC, software license, workflow packs	Hardware ₹30 lakh–₹55 lakh per production cell; installation and validation ₹2 lakh–₹6 lakh; annual AMC 10%–15% of hardware value; software ₹6 lakh–₹12 lakh per year	With deployed direct cost around ₹12 lakh–₹18 lakh, year-one blended GM can reach roughly 45%–58% if pricing discipline holds. For a normalized 2,000-samples/day lab, economic payback is typically 18–30 months unless throughput gains are strong	Best near-term cash profile if payment milestones are structured around PO, FAT, SAT, and acceptance. Working-capital need is moderate, mainly for inventory and receivables	Fits private labs that already buy analyzers and benches. Public procurement can be slower, often tender-driven, and may demand GeM-style or centralized procurement structures. Validation paperwork, IQ/OQ-style support, and SOP changes are mandatory in practice	Long enterprise cycles, price pressure, lumpy revenue, weak recurring base, expansion dependent on new hardware sales

Model family	Revenue streams	Indicative pricing mechanics in INR	Unit economics and buyer payback	Cash-flow profile and capital need	Adoption, procurement, and regulatory constraints	Risk vectors
RaaS / Opex subscription	Monthly subscription covering hardware use, support, software, monitoring, upgrades, optional minimum-uptime SLA	Deployment fee ₹3 lakh–₹8 lakh plus ₹2.5 lakh–₹6.5 lakh per month over 36–60 months, depending on scope and uptime guarantees	Gross margin is weaker early because the startup carries asset cost up front; steady-state GM can reach 55%–70% if utilization is high and service load is controlled. Buyer payback is often 3–9 months because upfront spend is low	Hardest on startup balance sheet. Requires financing line, asset SPV, or investor tolerance for negative hardware cash cycle	Very attractive for private buyers that want Opex approval. Less straightforward in public or hospital procurement unless managed services are clearly tenderable. Data-hosting and uptime clauses become major negotiation points	Underpriced contracts, uptime liabilities, hardware damage risk, renewal risk, poor asset recovery if a customer defaults
Hybrid hardware plus SaaS plus consumables	Upfront hardware, recurring orchestration software, support, PM kits, wear parts, workflow libraries, optional add-on modules	Hardware ₹18 lakh–₹30 lakh for entry deployment or pilot conversion; software and support ₹0.75 lakh–₹2 lakh per month ; PM kits and wear parts ₹1 lakh–₹4 lakh per year	Best balance of margin and retention. Blended GM can reach 55%–68% over time because recurring software and support are high-margin relative to hardware. Buyer payback for a normalized lab is often 9–18 months	Better than RaaS because customer co-funds hardware. Better recurring economics than pure CapEx. Capital need remains manageable if upfront collections cover most manufacturing cost	Easiest to explain to private chains and reference labs: “equipment plus software plus service.” Consumables are less powerful here than in analyzers, so they should be positioned as PM kits and standardized accessories rather than as the core annuity	Risk that customers try to strip out software or support after initial deployment; need strong technical lock-in through orchestration, dashboards, and analytics

Model family	Revenue streams	Indicative pricing mechanics in INR	Unit economics and buyer payback	Cash-flow profile and capital need	Adoption, procurement, and regulatory constraints	Risk vectors
Transactional per-test pricing	Fee per test, per sample automated, or per successful routed handoff; often bundled with minimum monthly floor	Usually needs to stay low, roughly ₹1.5–₹5 per test or ₹3–₹12 per automated handoff , because public retail test prices often range from about ₹270 to ₹1,800 for many common tests and realized B2B economics are lower	Can look attractive to customers, but monetization can be too thin unless volumes are very high. Startup GM is heavily utilization-sensitive. Buyer payback is immediate because spend becomes variable	Poor startup cash profile unless protected by minimum guarantees. Hardware must still be financed before usage ramps	Hard to define the billable unit in diagnostics because one sample can generate multiple tests and multiple bench movements. Requires auditable logs and LIS-level counting with dispute resolution	Volume volatility, attribution disputes, underpricing, misalignment between “test count” and “automation value”
Revenue-share with labs	Share of incremental revenue, TAT-linked upsell, or documented labor savings; often with minimum monthly guarantee	Typical structure would need a floor of ₹1.5 lakh–₹3 lakh per month plus a share, for example 3%–8% of agreed incremental revenue or 15%–30% of audited savings	Can produce strong ROI narratives for the customer, but startup economics are noisy and collection quality depends on baseline measurement and trust. Buyer payback is immediate if the floor is low	Weak near-term cash flow because the startup still funds deployment while upside is delayed and audited later. Capital need resembles RaaS, but with more uncertainty	Procurement teams often dislike ambiguous formulas. Finance teams will challenge baseline definitions and causality. Hardest model to take through formal tendering or chain-wide rollout	Measurement disputes, manipulation of baseline, delayed collections, low transparency, weak bankability for debt financing

Model family	Revenue streams	Indicative pricing mechanics in INR	Unit economics and buyer payback	Cash-flow profile and capital need	Adoption, procurement, and regulatory constraints	Risk vectors
Licensing and IP	OEM license, orchestration software license, driver or workflow IP, NRE, white-label integrations, developer tool access	NRE ₹20 lakh–₹75 lakh per integration; annual license ₹50 lakh–₹2 crore or more per OEM or chain partner, depending on scope	Direct GM can be 75%–90% once the platform is mature. Customer payback is partner-specific and not comparable on a per-lab basis	Low deployment capital but high product and QA investment. Cash flow depends on few large deals with long sales cycles	Strong fit only after the company has credible fleet data, stable drivers, proven integrations, and protectable IP. OEM and regulated-environment diligence is heavy	Concentration risk, long BD cycles, dependency on partner roadmap, IP leakage, lower near-term control of end customer

Indicative model, not price quotes. Public anchors include Opentrons list pricing and services, Automata's cloud/support and integration posture, Indian hospital automation benchmarks at SGPGIMS and AIIMS Bhopal, official quality and tracking claims from large Indian diagnostics chains, and current INR conversion. ¹³

A few conclusions follow from the table. First, **CapEx** is strongest where the lab already has a robust equipment-buying process and wants the asset on its own books; it is weakest where the startup needs recurring revenue and predictable fleet economics. Second, **RaaS** removes customer friction but transfers financing and uptime risk to the vendor; it works best when the lab is strategically valuable enough to justify that burden. Third, the **hybrid model** wins because it allows the startup to recover most hardware cash up front while still building durable annual revenue and operational control through software, dashboards, workflow libraries, and support. ¹⁴

Fourth, **transactional and revenue-share models are structurally weaker for phase-one diagnostics automation than they are in categories like assay reagents or payments infrastructure**. Metropolis' public test catalog shows that many routine tests sit in the low-hundreds to low-thousands of rupees range, which means an automation vendor's fee must usually remain a small share of lab economics. That is precisely why analyzer-style reagent-rental models are common: the vendor owns recurring chemistry revenue. A pre/post-analytical startup generally does not, so purely usage-linked pricing tends to compress monetization unless paired with a minimum monthly commitment. ¹⁵

Fifth, **licensing and IP are real but delayed**. Opentrons' OEM and integrated-device positioning shows that automation platforms can expand into partner-driven revenue streams, integrated workflows, and white-labeled or embedded control layers. But those deals are strongest when the underlying platform already has credible deployments, custom drivers, and proven partner integrations. For this startup, licensing should be treated as a post-beachhead expansion, not as the main way to reach initial revenue. ¹⁶

The KPI summary below normalizes economics to a **midpoint profile of about 2,000 samples per day and 26 operating days per month**. For lumpy models such as CapEx and Hybrid, “ARPU” should be read as **recognized revenue in year one**, not recurring ARR alone.

Model family	Normalized year-one ARPU or equivalent	CAC assumption	CAC payback to startup	Buyer payback	Five-year LTV gross profit	Annual logo churn assumption	Steady-state gross margin
CapEx hardware sale	₹33 lakh–₹61 lakh in year one; then ₹3 lakh–₹8 lakh annual support/software	₹6 lakh–₹12 lakh	1–4 months after close	18–30 months	₹18 lakh–₹35 lakh	<5% on installed base, but repeat-sale cycle is long	45%–58% blended year one
RaaS / Opex	₹30 lakh–₹78 lakh per year	₹5 lakh–₹9 lakh	8–14 months	3–9 months	₹35 lakh–₹90 lakh	8%–15%	55%–70% at scale
Hybrid	₹28 lakh–₹52 lakh in year one; ₹10 lakh–₹28 lakh recurring from year two	₹5 lakh–₹10 lakh	6–12 months	9–18 months	₹30 lakh–₹65 lakh	5%–10%	55%–68%
Transactional per-test	₹12.5 lakh–₹25 lakh at ₹2–₹4 per automated test-event equivalent	₹4 lakh–₹8 lakh	12–24 months unless protected by a floor	0–6 months	₹20 lakh–₹45 lakh	7%–15%	35%–60%
Revenue-share	₹15 lakh–₹30 lakh if protected by a minimum and a modest share	₹6 lakh–₹10 lakh	12–24 months	Immediate if floor is modest	₹18 lakh–₹50 lakh	10%–20%	30%–55%

Model family	Normalized year-one ARPU or equivalent	CAC assumption	CAC payback to startup	Buyer payback	Five- year LTV gross profit	Annual logo churn assumption	Steady- state gross margin
Licensing and IP	Not comparable per lab; roughly ₹50 lakh– ₹200 lakh plus per partner	₹10 lakh– ₹25 lakh	18–36 months	Partner- specific	₹75 lakh– ₹300 lakh	Deal concentration risk, not classic churn	75%– 90%

Modeled assumptions by the author, using the same public anchors as the main business-model table. ¹⁷

Pilot go-to-market options

The startup should run **paid pilots**, not free tech showcases. In regulated and accreditation-sensitive lab environments, pilots do not only prove hardware; they force the real commercial work to happen: SOP alignment, IT integration, quality signoff, training cadence, floor ownership, and escalation paths. Opentrons and Automata both publicly frame installation, IQ/OQ or acceptance testing, protocol development, and ongoing support as core parts of deployment, not as afterthoughts. That is a good signal for how to contract pilots in diagnostics. ¹⁸

Pilot option	Target customer profile	Why it is attractive	Likely sales cycle	Suggested pilot terms	Pilot success metrics	Best conversion path
Regional private reference lab	Independent or promoter-led lab, roughly 1,500–3,000 samples/day , existing LIS, multi-bench workflow, local decision-maker	Fastest commercial learning. Decision-making is usually tighter than in large chains, and pain is immediate because manual glue work is visible on the floor	6–12 weeks for pilot approval; 2–4 months to full contract	12-week paid pilot at ₹3 lakh–₹5 lakh; customer provides space, operator champion, and LIS access; pilot fee converts into purchase credit on rollout	≥70% automated routing in scoped workflow, ≥98% uptime during staffed hours, >50% reduction in manual touches per sample on scoped lane, measurable p95 TAT improvement, near-zero mis-sort events	Hybrid default
Chain hub or spoke-hub central lab	Large diagnostics chain hub, roughly 3,000–5,000 samples/day , national quality function, multiple analyzer vendors, strong brand value	Best lighthouse value and highest expansion upside if one site succeeds. Also the strongest source of fleet data and workflow templates	3–6 months for pilot approval; 6–9 months for full rollout	4–6 month paid pilot at ₹5 lakh–₹8 lakh, with explicit co-development boundaries, reference rights, and pre-agreed multi-site pricing schedule	98.5%+ uptime, 15%–25% p95 TAT reduction on scoped workflow, 2–4 FTE-equivalent redeployment, complete audit trail, analyzer/LIS integration stability, supervisor adoption	Selective RaaS or Hybrid, depending chain budget structure

Pilot option	Target customer profile	Why it is attractive	Likely sales cycle	Suggested pilot terms	Pilot success metrics	Best conversion path
Hospital enterprise or PPP or public hub	Tertiary-care hospital lab, district integrated lab, or public-private diagnostic hub, roughly 800–2,000 samples/day but high urgency and service complexity	Good for credibility, tenderability, and public-health use cases, but also the slowest path because procurement and compliance are heavy	6–12 months or longer	Managed-service style pilot or innovation procurement; ideally 6 months with staged milestones and documentary acceptance criteria	STAT TAT improvement, chain-of-custody completeness, quality-team acceptance, validated SOP package, training completion, no audit-critical failures	Hybrid later, public managed service only when procurement path is clear

The right **pilot metric set** should be operational, financial, and quality-linked. The most important ones are not “robot cycles completed.” They are: reduction in manual touches per sample, automation coverage on the scoped workflow, barcode-read success rate, downtime minutes per week, mis-sort or exception rate, rerun burden, p95 TAT on target benches, and FTE hours redeployed rather than removed. That framing matters because procurement teams, quality teams, and lab managers all care about different proof points.

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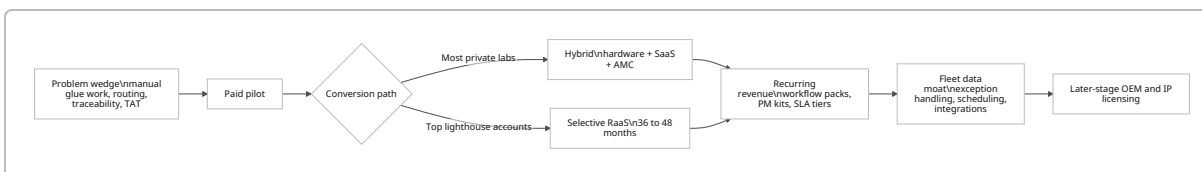
A strong standard pilot contract should include four commercial protections. First, a definition of the **billable workflow scope**. Second, clear **customer obligations** around bench space, utilities, LIS access, and a local process owner. Third, a **conversion clause** that credits 50% to 100% of pilot fees against a production deployment. Fourth, a **data-rights clause** that lets the startup use de-identified operational events to improve scheduling, vision, and exception handling. Without those protections, the startup may create buyer education and product learning without monetizing either. That risk is particularly high if the company drifts toward unpaid “innovation pilots.” [filecite turn0file1](#)

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Recommended commercialization path

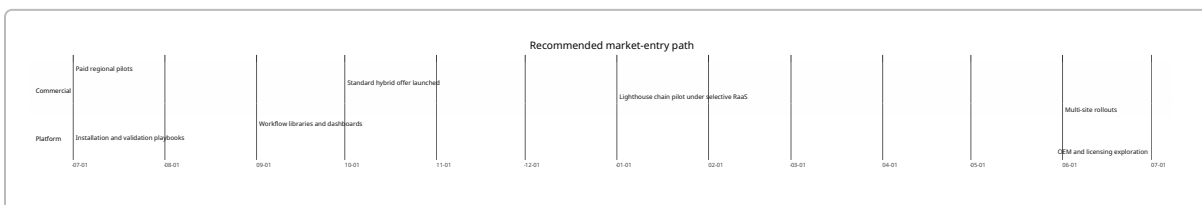
The recommended sequence is straightforward: **Hybrid as the standard commercial product, selective RaaS for one to three lighthouse deployments, transactional or revenue-share only as overlays with minimum guarantees, and licensing or OEM deals only after the installed base is real**. That sequence fits the operational wedge in your pitch deck, addresses the GTM and unit-economics gap called out in the validation report, and aligns with how serious lab-automation vendors already monetize software, services, integrations, and regulated-environment support. [filecite turn0file1](#) [filecite turn0file2](#)

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The most practical **price book** for the next 12 to 18 months would look like this. The **standard production offer** should be a hybrid bundle: **₹22 lakh to ₹35 lakh** for a narrow production deployment or **₹30 lakh to ₹45 lakh** for a broader version, plus **₹0.8 lakh to ₹1.5 lakh per month** for software, monitoring, and support, plus annual PM or wear-part revenue. The **lighthouse managed-service offer** should be reserved for the most referenceable accounts and priced at **₹2.5 lakh to ₹4.5 lakh per month**, ideally with a setup fee and a 36- to 48-month term. The company should **not** default to low-fee per-test pricing unless the contract includes a floor that covers gross profit and financing cost. ²²

That recommendation is also strategically important because the startup needs a future revenue stack that compounds. Hardware alone does not compound fast enough. A hybrid model creates three assets simultaneously: deployment revenue, recurring software and support, and field data that later justifies workflow libraries, exception-prediction features, and licensing conversations. Opentrons and Automata both make visible how modern automation companies accumulate value above the hardware layer through cloud control, workflow creation, integration support, and OEM pathways. That is the model to emulate, adapted to Indian diagnostics economics rather than imported as-is. ²³



Open questions and limitations

This report is rigorous on commercial structure, but a few items remain inherently uncertain and should be treated that way.

- The **BOM and deployed-cost stack** is a modeled assumption anchored to public comparables, not to supplier quotations for your exact motion system, safety architecture, vision stack, or local assembly approach. Real vendor quotes could move those numbers materially.
- The **best regulatory classification path** depends on product claims. If the system is positioned as pre/post-analytical automation and traceability infrastructure, the path is different than if it begins making diagnostic claims or becomes tightly embedded in an IVD decision chain. This report therefore treats regulation mainly as a procurement-and-validation burden rather than giving formal legal classification advice.
- The **per-test pricing discussion** is anchored partly to public retail test prices at Metropolis, which are useful as a visible market ceiling but are not the same as every lab's realized B2B economics.

Transactional pricing should therefore be validated against actual customer mix before productizing it. ²⁴

- **Public procurement** for hospital and government customers is structurally slower and less uniform than private procurement. GeM-linked procurement norms exist, and some state hospital systems have tightened controls over direct purchasing, but actual route-to-market still varies by institution and state. ²⁵

On balance, the evidence supports a clear answer: **sell the phase-one product as focused operational infrastructure with recurring software and service, not as a one-off robot and not as a thin per-test utility**. That is the model most consistent with the market, the product wedge, and the current maturity of the concept. ¹ ⁶ ¹² ¹³ ¹⁴ ¹⁵ ¹⁶ ¹⁷ ¹⁸ ¹⁹ ²⁰ ²¹ ²² ²³ ²⁴ ²⁵

¹ ⁶ ¹² <https://timesofindia.indiatimes.com/city/bhopal/aiims-bhopals-diagnostic-system-to-boost-patient-care/articleshow/124199566.cms>

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